Input Power

File: input_power.kicad_sch

5V Power Supply

File: 5v_power_supply.kicad_sch

Input Signal

File: input_signal.kicad_sch

Microcontroller

File: microcontroller.kicad_sch

USB UART Bridge

File: USB_UART_Bridge.kicad_sch

RGB LED

File: rgb_led.kicad_sch

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Sheet: /

File: HEI_Shift_Light.kicad_sch

Title: HEI Shift Light

Size: A	Date: 2026-05-29	**Rev: A**
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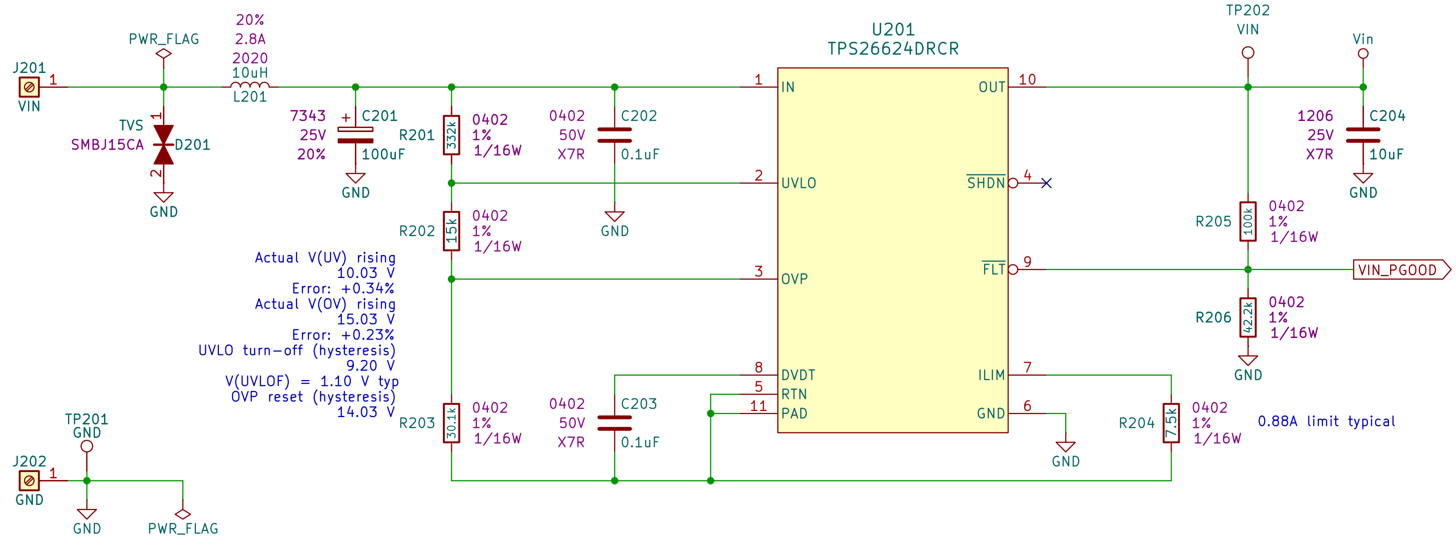
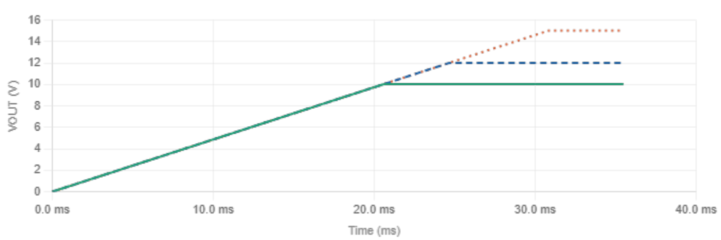


DVDT cap:
Min corner: I=2.33 µA, GAIN=23.9 V/V (longest ramp)
Typ: I=1.98 µA, GAIN=24.6 V/V
Max corner: I=1.68 µA, GAIN=25.2 V/V (shortest ramp – slowest I, widest gain)

VOUT target	Rise time – min	Rise time – typ	Rise time – max	Slew rate (typ)
10.03 V (UVLO threshold)	18.0 ms	20.6 ms	23.7 ms	0.49 V/ms
12 V (Nominal)	21.5 ms	24.6 ms	28.3 ms	0.49 V/ms
15.03 V (OVP threshold)	27.0 ms	30.9 ms	35.5 ms	0.49 V/ms



VIN_PGOOD divider:
VEN at VUV (min VOUT)
2.98 V
Must be ≥ 0.65 V □
VEN at VOV (max VOUT)
4.46 V
□ Within safe range
Divider current at VUV
70.53 µA
Keep ≥ 10× EN leakage
□ V(EN) min = 2.98 V (need ≥ 0.65 V)□ V(EN) max = 4.46 V□ Fault: FLT pulls EN to ~0 V (< 0.3 V)

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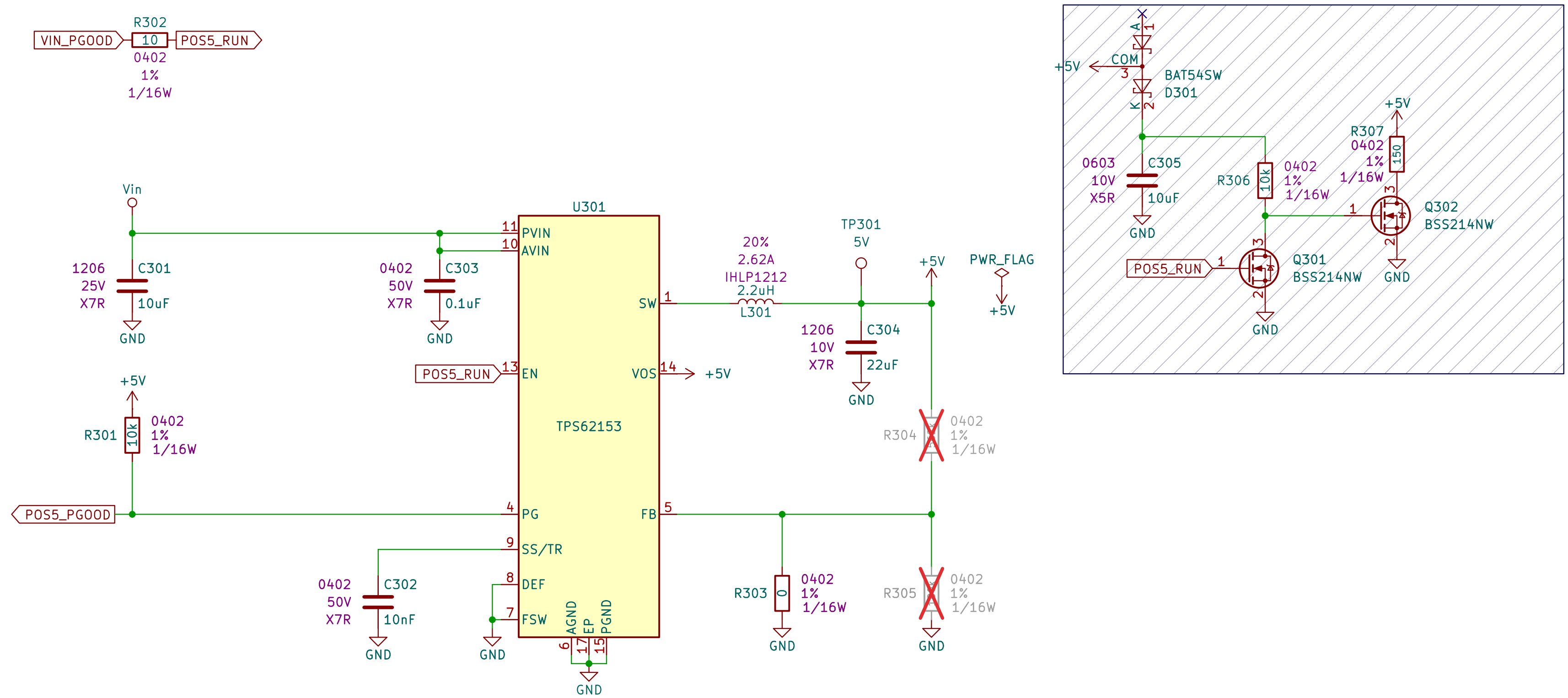
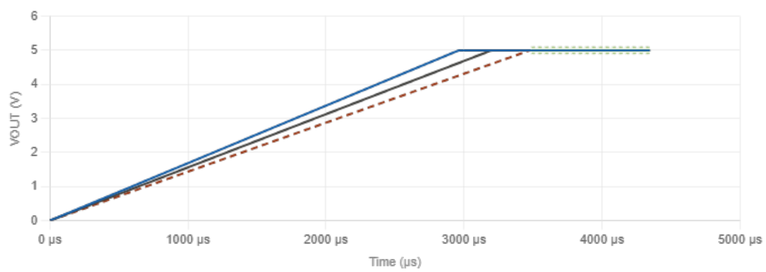
Get rid of discharge? Do something more simple?



Rise times & settled output range

Min time: I=2.7 µA (fastest) | Typ: I=2.5 µA | Max time: I=2.3 µA (slowest)

Condition Rise time – min Rise time – typRise time – max Settled VOUT
DEF=low (nominal 5.00 V) 2.963 ms3.200 ms3.478 ms4.910 – 5.090 V
DEF=high (+5% = 5.25 V) 2.963 ms3.200 ms3.478 ms5.155 – 5.345 V



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Sheet: /5V Power Supply/
File: 5v_power_supply.kicad_sch
Title: HEI Shift Light
Size: A Date: 2026-05-29 **Rev: A**
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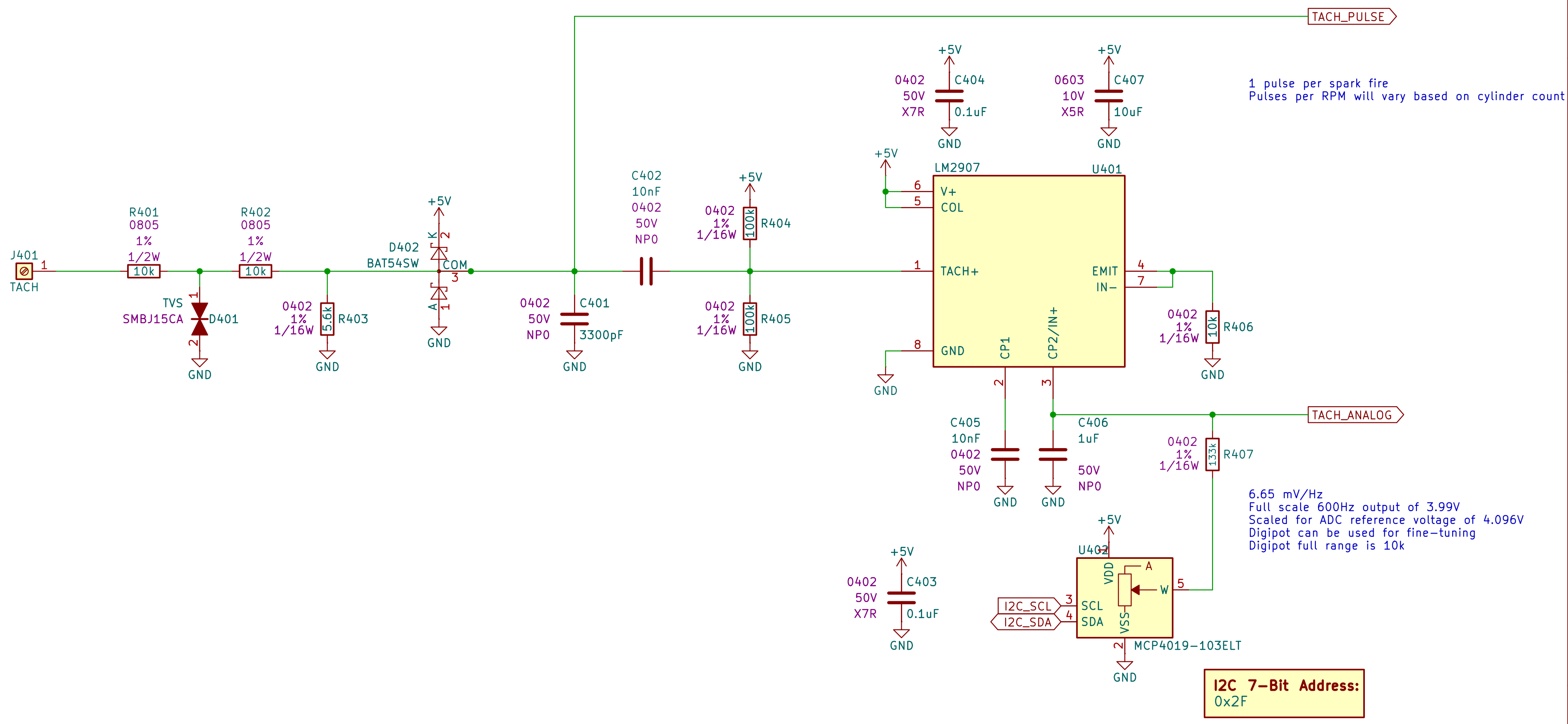

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Sheet: /Input Signal/
File: input_signal.kicad_sch

Title: HEI Shift Light

Size: A	Date: 2026-05-29	**Rev: A**
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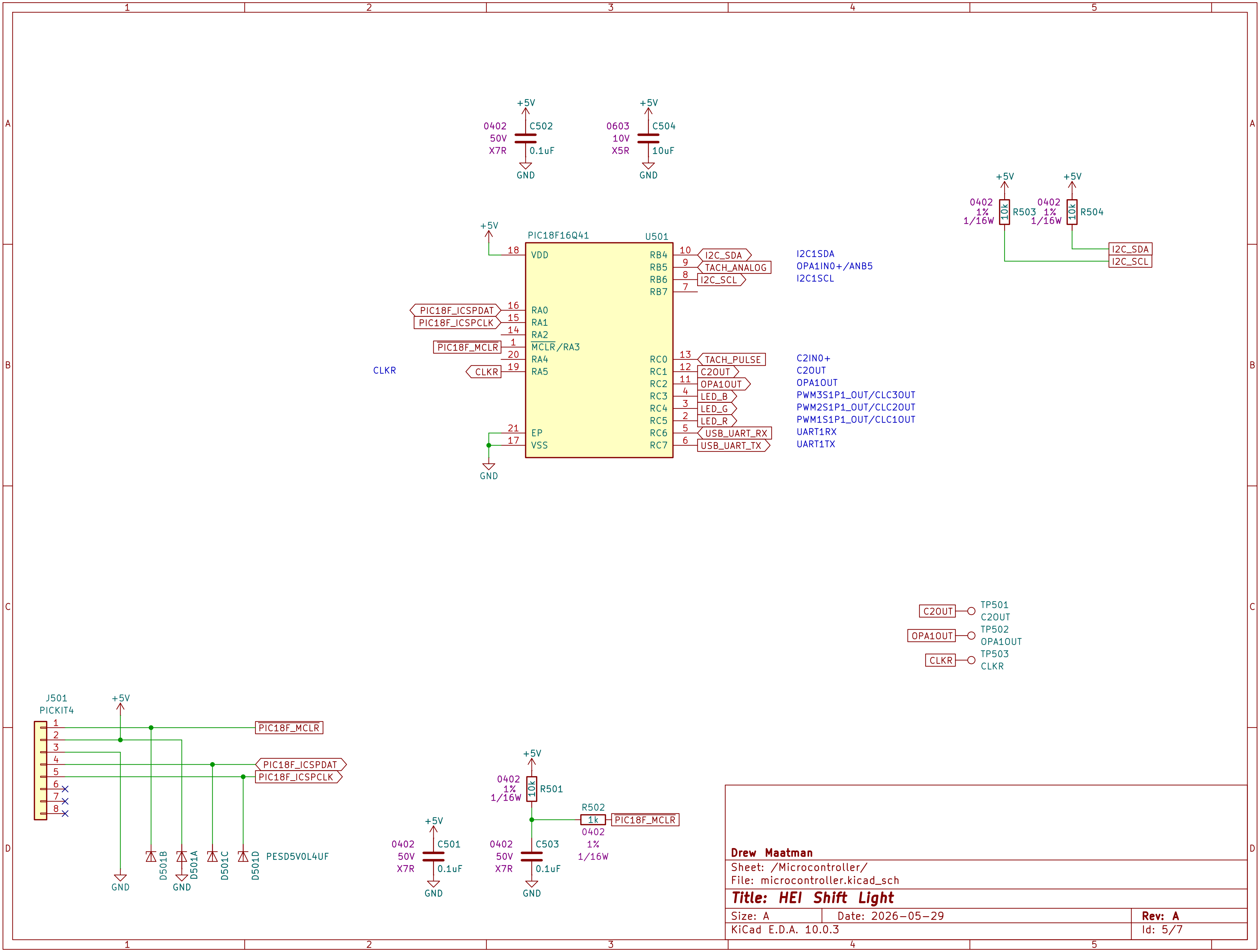
+5V
C502
0402
50V
X7R
0.1uF
GND
C504
0603
10V
X5R
10uF
PIC18F16Q41
U501
VDD
RB4
RB5
RB6
RB7
RA0
RA1
RA2
MCLR/RA3
RA4
RA5
RC0
RC1
RC2
RC3
RC4
RC5
RC6
RC7
EP
VSS
I2C_SDA
TACH_ANALOG
I2C_SCL
PIC18F_ICSPDAT
PIC18F_ICSPCLK
PIC18F_MCLR
CLKR
TACH_PULSE
C2OUT
OPA1OUT
LED_B
LED_G
LED_R
USB_UART_RX
USB_UART_TX
I2C1SDA
OPA1IN0+/ANB5
I2C1SCL
C2IN0+
C2OUT
OPA1OUT
PWM3S1P1_OUT/CLC3OUT
PWM2S1P1_OUT/CLC2OUT
PWM1S1P1_OUT/CLC1OUT
UART1RX
UART1TX
R503
R504
10k
0402
1%
1/16W
TP501
TP502
TP503
J501
PICKIT4
D501A
D501B
D501C
D501D
PESD5V0L4UF
R501
R502
1k
C501
C503
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Sheet: /Microcontroller/
File: microcontroller.kicad_sch
Title: HEI Shift Light
Size: A
Date: 2026-05-29
Rev: A
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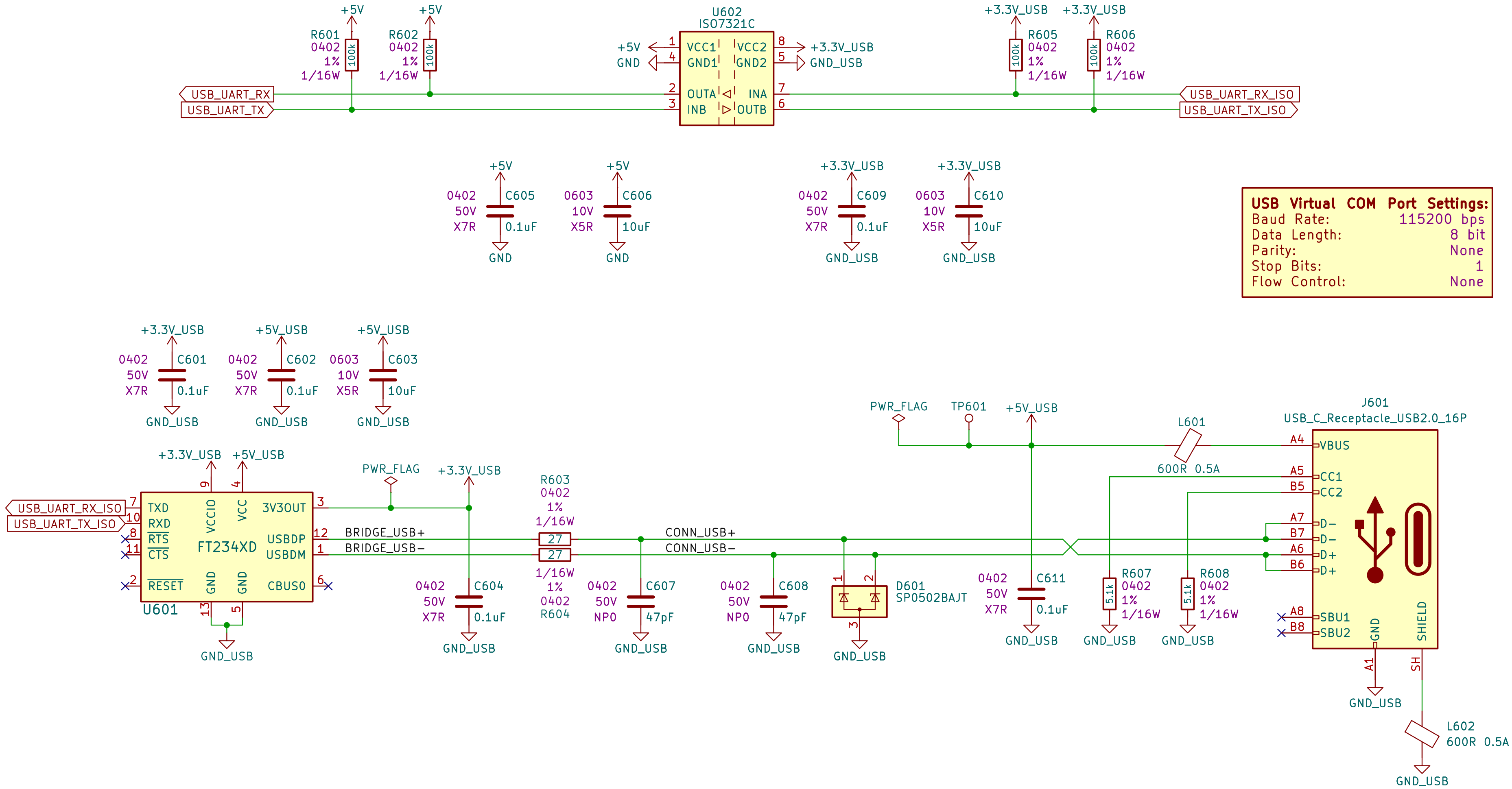

USB Virtual COM Port Settings:	
Baud Rate:	115200 bps
Data Length:	8 bit
Parity:	None
Stop Bits:	1
Flow Control:	None

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Sheet: /USB UART Bridge/

File: USB_UART_Bridge.kicad_sch

Title: HEI Shift Light

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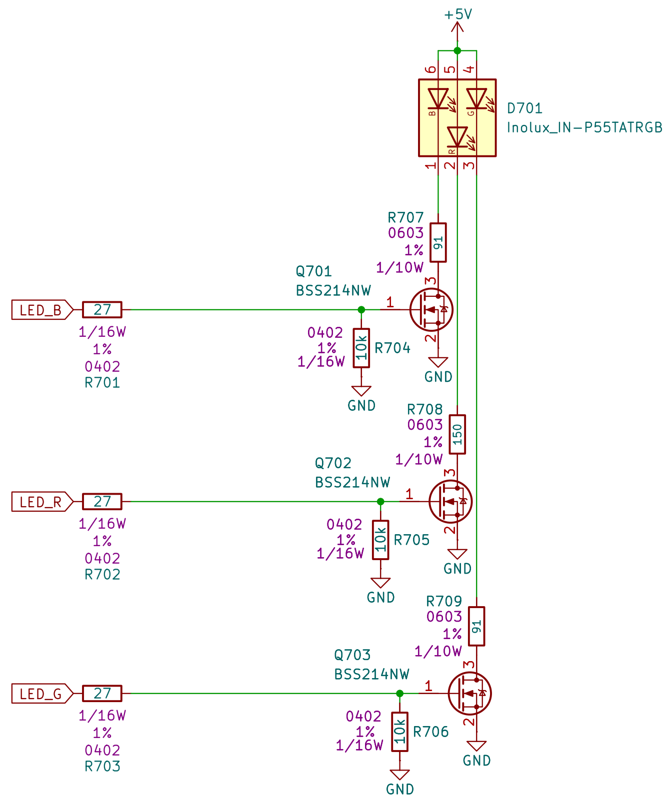

Based on the datasheet, the typical forward current (I_F) for testing is 20mA,
with an absolute maximum of 25mA.
For "maximum brightness" while maintaining reliability, I recommend targeting 20mA.

Using Ohm's Law (R = (VCC - VF)*IF with VCC = 5V:

Channel	Typ Vf	Calculation (20mA)	Recommended Res.	Max Bright (25mA)
Red	2.0V	(5 - 2.0) / 0.020	150 Ohm	120 Ohm
Green	3.2V	(5 - 3.2) / 0.020	91 Ohm (Standard)	72-75 Ohm
Blue	3.2V	(5 - 3.2) / 0.020	91 Ohm (Standard)	72-75 Ohm

Design Notes:
- Power Dissipation: At 20mA, the resistors dissipate approx 36-60mW.
  Standard 0603 or 0805 SMD resistors are more than sufficient.
- Safety: I strongly recommend the 20mA values. Pushing to 25mA provides
  a marginal increase in brightness but reduces the LED's lifespan and increases heat.

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Sheet: /RGB LED/
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